

Solution Requirements

Date	28 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users/Admin log in securely to access the flood prediction system.	High
2	Authorization Levels	Admin can manage datasets and models. Users can enter weather parameters and view predictions.	High
3	External Interfaces	Uses historical flood/weather dataset, Flask web application, and trained ML model (.pkl).	High
4	Transaction Processing	User enters rainfall and weather parameters; the system processes inputs and generates flood prediction instantly.	High
5	Reporting	Displays prediction result (Flood/No Flood) with risk level on the dashboard.	Medium
6	Business Rules	Prediction is generated only after validating all required input values. Missing or invalid data is rejected.	High
7	Compliance to Laws or Regulations	User information is kept confidential and the application follows general data privacy practices.	Medium
8	Other	Supports future model updates and additional weather parameters without affecting existing functionality.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Generate prediction quickly after user input.	Response time less than 2 seconds
2	Scalability	Able to support increasing users and larger datasets.	Supports 500+ concurrent users
3	Security & Data Privacy	Protect user input and prevent unauthorized access.	Secure authentication and encrypted communication
4	Reliability & Availability	System should provide continuous service with minimal downtime.	99% uptime
5	Usability & Accessibility	Simple, responsive, and easy-to-use interface.	Mobile-friendly and intuitive UI
6	Other	Maintainable, portable, and easy to update with new ML models.	Modular code structure and easy deployment